

TECHNICAL CASE STUDY

Customer Behavior & Revenue Analytics

An end-to-end data pipeline transitioning raw ledger logs to production-ready enterprise intelligence.

Data Analytics & Engineering Portfolio
Project Deployment: June 2026



The Operational Challenge

Bridging Siloed Data Structures

- ❗ **Siloed Core Logs:** Unorganized retail transactions block clear lifecycle analysis and slow strategic optimization.
- 🗄️ **Database Demands:** 3,900 distinct consumer transaction records requiring structural validation.
- 🎯 **Execution Target:** Programmatically transition raw flat-file records into an optimized transactional model.



Key Metrics at a Glance

3.9K

TOTAL CUSTOMERS

Active unique customer transactional records analyzed across regions.

\$59.76

AVG PURCHASE

Baseline transactional size recorded per consumer ledger line.

3.75

AVG REVIEW

Overall satisfaction rating captured across product inventories.

Pipeline Architecture

A multi-tiered, modular data architecture engineered to guarantee scalable processing and analytical accuracy.



1. Ingestion

Parsing dirty CSV records directly from transactional endpoints.



2. Python ETL

Programmatically cleansing schemas and validating datatype constraints.



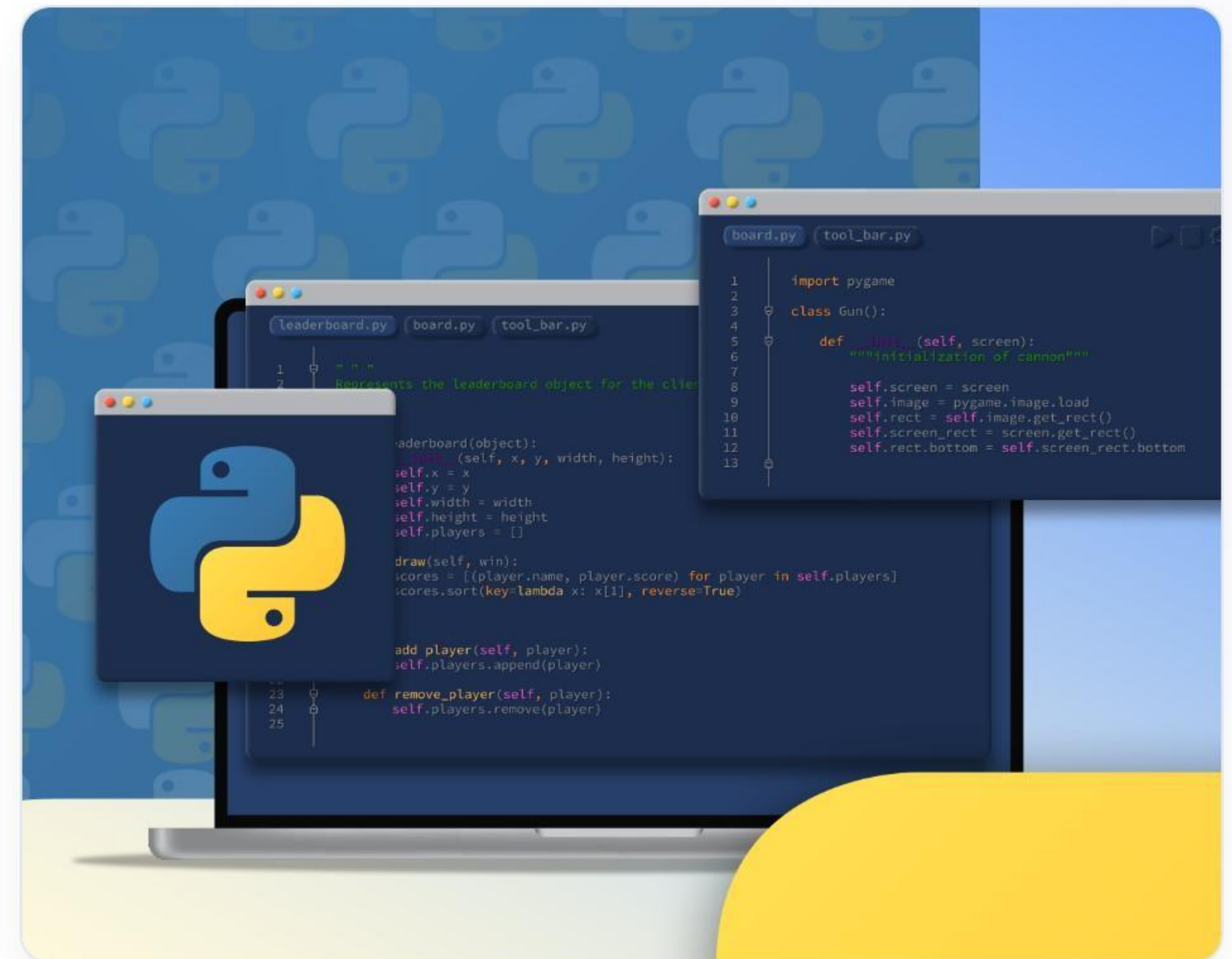
3. PostgreSQL

Structuring analytics-ready databases using optimized SQL schemas.

Phase A: Python ETL Engineering

Validating Data Integrity

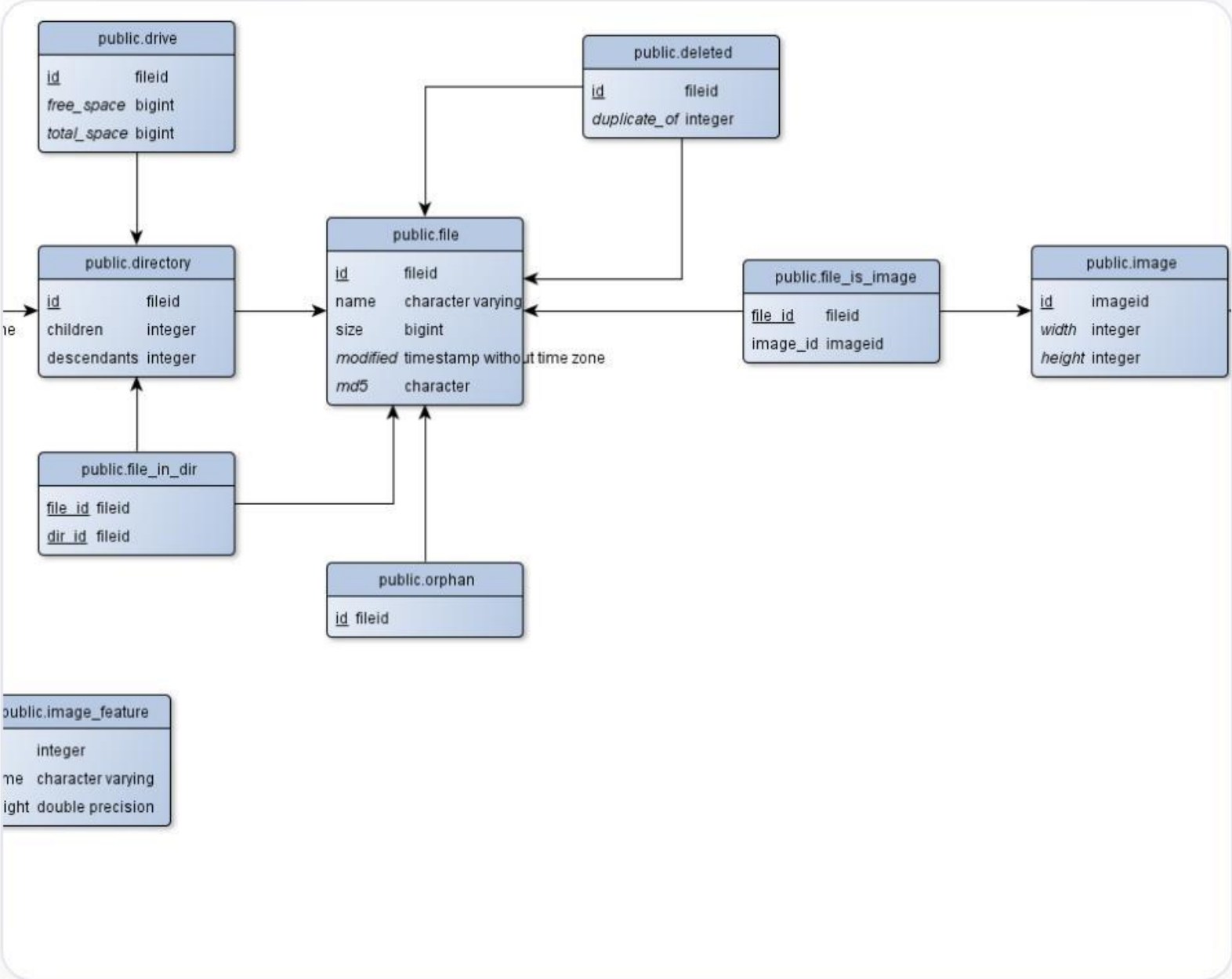
-  **Schema Verification:** Systematically casting object types using Pandas to ensure absolute database ingestion stability.
-  **Constraint Auditing:** Programmatically verifying matrix fields to enforce clean structures with zero null records.
-  **Automated Serialization:** Connecting Python natively to PostgreSQL using SQLAlchemy for safe transactional inserts.



Phase B: Database Storage

Structuring Relational Models

-  **Centralized Schemas:** Modeled cleanly in pgAdmin to minimize storage footprint and database redundancy.
-  **Native Window Functions:** Utilizing queries like DENSE_RANK() to build customer hierarchy maps natively in the SQL engine.
-  **Latency Reductions:** Optimizing structural execution scripts to run lightning-fast updates for the reporting layer.



Executive Dashboard

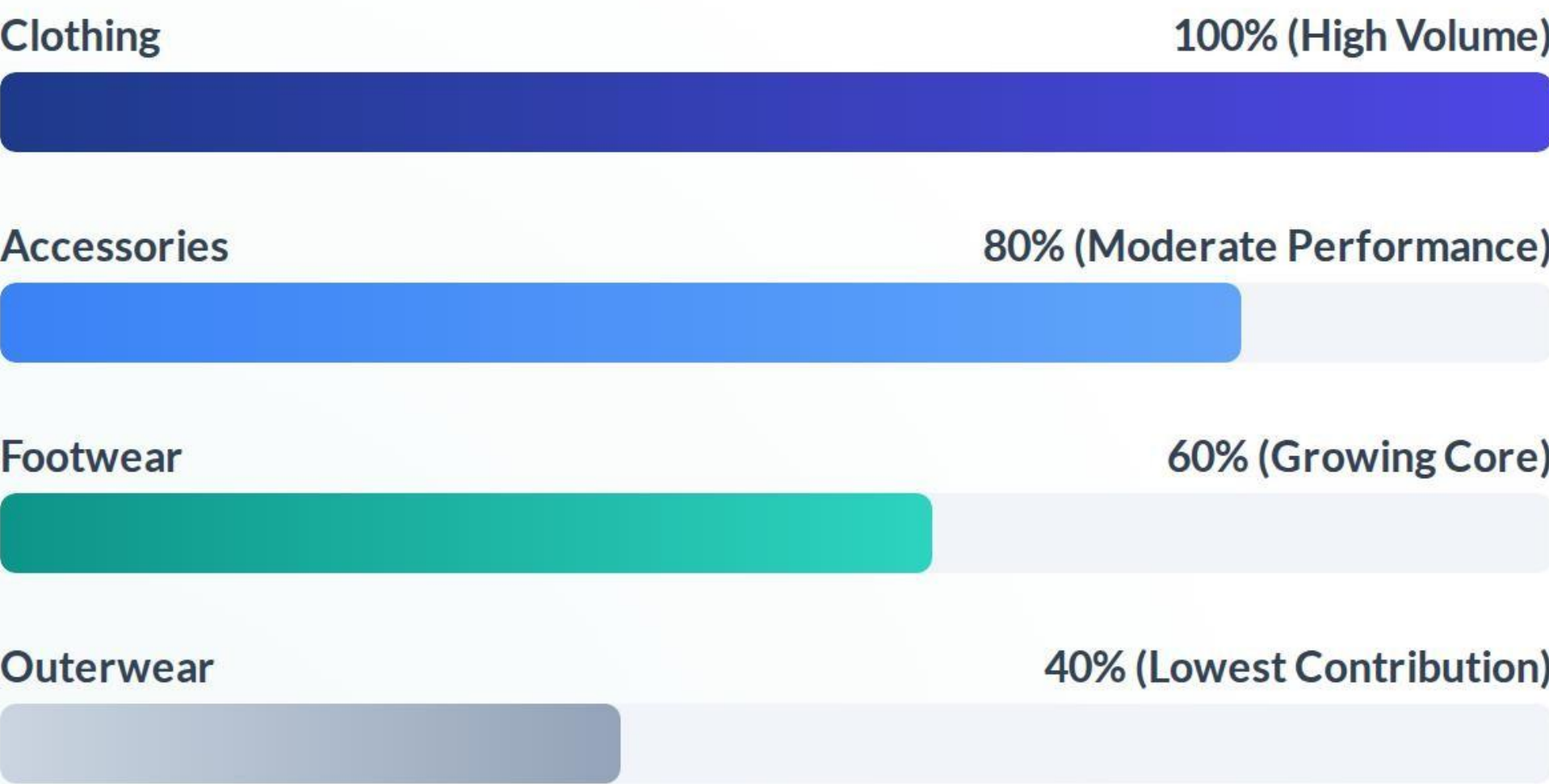
Translating Data to Strategy

An interactive business intelligence application designed to expose and track customer transactions in real-time.

Enables the executive team to slice KPIs dynamically based on Subscription Status, Demographics, Logistics, and Product Categories, stripping out unneeded complexity.



Category Revenue & Sales



Category Insights

Clothing remains the absolute anchor of transactional volume and revenue metrics.

Conversely, **Outerwear** sits at the tail end of sales sheets, highlighting a tactical consolidation or targeted marketing opportunity.

Subscription & Age Insights



Active Subscribers (27%)

- Represents high customer lifetime value, offering a direct structural retention anchor.

Non-Subscribed (73%)

- An outstanding conversion window for loyalty program outreach campaigns.

A Fully Integrated Data Pipeline

- .  **Python ETL Engine:** Delivers the automated dataset verification and cleaning routines.
- .  **PostgreSQL Storage:** Operates as the high-performance relational storage schema.
- .  **Reporting Platform:** Translates underlying complex tables into clear management actions.

Transitioning unorganized shopping logs to tactical business decision